

Nikhil Singh, Fall 2026

COSC 3

Introduction to Machine learning

Reminders

Quiz 2 on Wednesday Oct. 7th

Assignment 2 due
Monday Oct. 12th at 11:59pm

Prediction from **features**

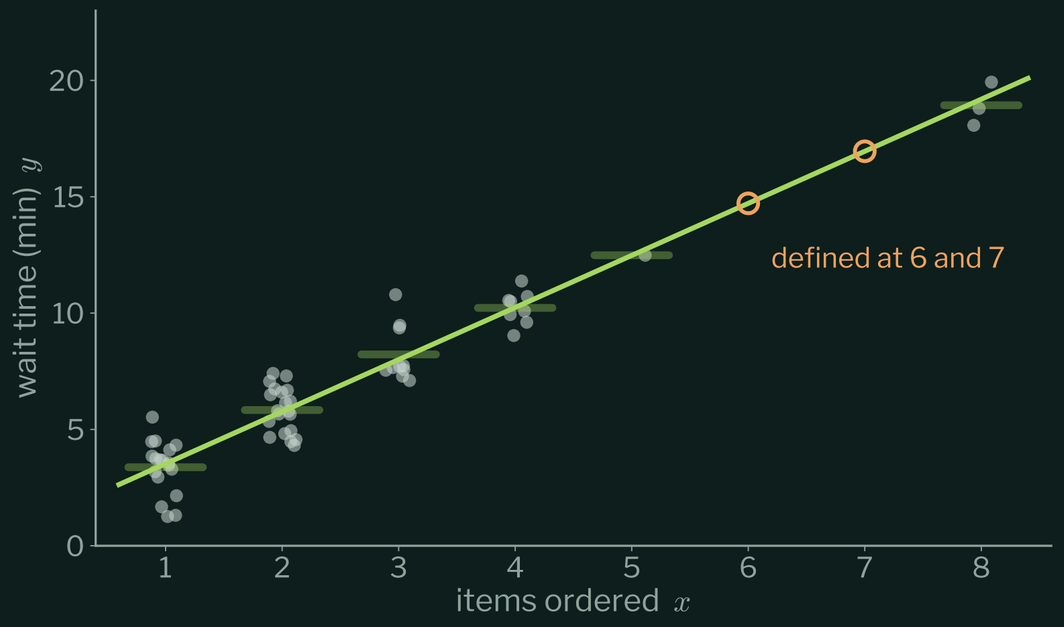
$$\hat{y} = c$$

Same for all cases

$$\hat{y} = f(x)$$

Depends on case-specific features x

Mean outcome as **line** in x



$$y = mx + b \quad \longrightarrow \quad \hat{y} = \theta_1 x + \theta_0$$

$\boldsymbol{\theta}$: parameters to learn

x: feature

θ_0: baseline offset when $x = 0$

θ_1: minutes per extra item

Prediction for every x, including 6–7

Residuals



$e_i = y_i - \hat{y}_i$

Vertical gap: point to line

$e_i > 0$: point above line

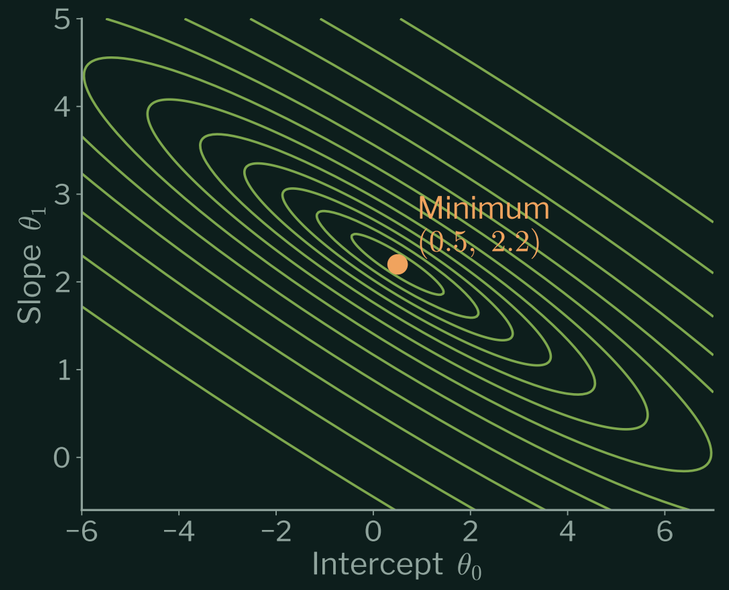
$e_i < 0$: point below line

Least-squares objective

$$\text{Before: MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - c)^2$$

$$\text{Now: } R(\theta_0, \theta_1) = \frac{1}{n}\sum_{i=1}^{n}(y_i - (\theta_1 x_i + \theta_0))^2$$

$$(\theta_0^*, \theta_1^*) = \arg\min_{\theta_0, \theta_1} R(\theta_0, \theta_1)$$



One-feature OLS formulas

Quantity	Formula	Equivalent	Units
Slope θ_1^*	$\frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$	$r\frac{s_y}{s_x}$	$[y]/[x]$
Intercept θ_0^*	$\bar{y} - \theta_1^*\bar{x}$	—	$[y]$

What if we replace MSE $\longrightarrow$ MAE?

$$x = (1, 2, 3, 4), \quad y = (3, 4, 8, 9), \quad \hat{y}_i = \theta_0 + \theta_1 x_i$$

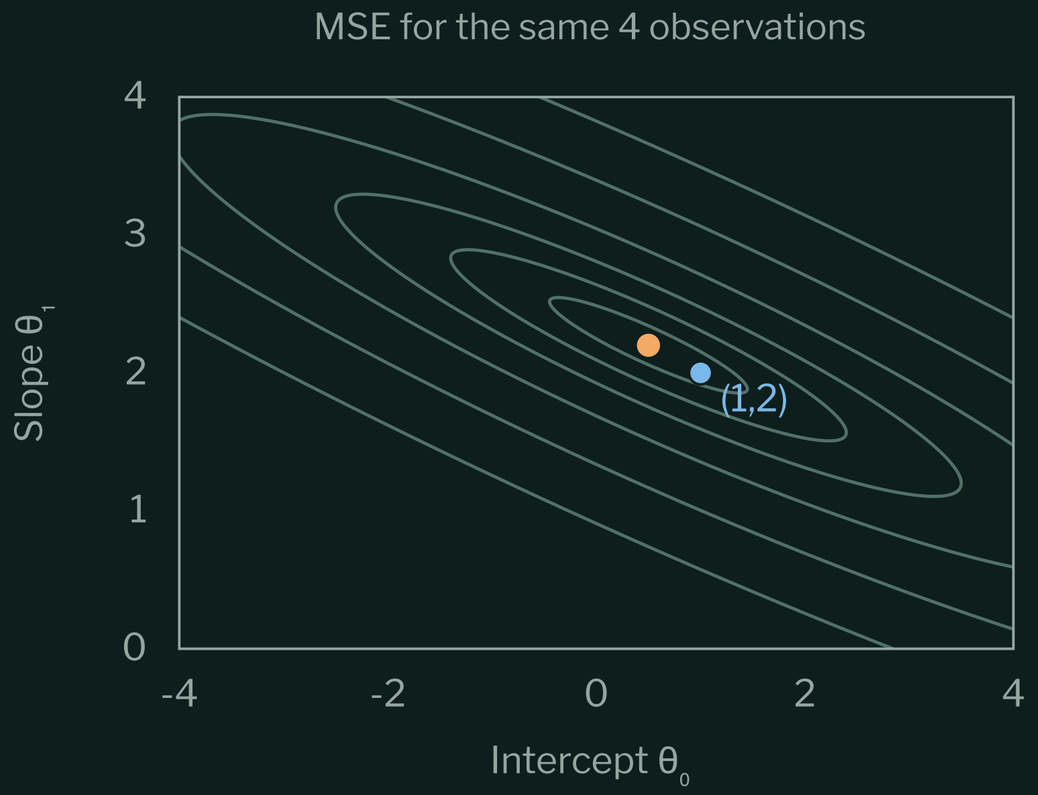
$$R_{\text{MSE}} = \frac{1}{4}\sum_i (y_i - \hat{y}_i)^2 \qquad R_{\text{MAE}} = \frac{1}{4}\sum_i |y_i - \hat{y}_i|$$

Candidate line	MSE	MAE
OLS: $\hat{y} = 0.5 + 2.2x$	0.45	0.60
Alternative: $\hat{y} = 1 + 2x$	0.50	0.50

OLS coefficients no longer minimize this

Can we repeat derivation and get closed-form solution?

Search over parameters of a fixed prediction rule



Blue points: tested candidates

Orange: OLS minimum at (0.5,2.2)

$x = (1, 2, 3, 4)$ and $y = (3, 4, 8, 9)$

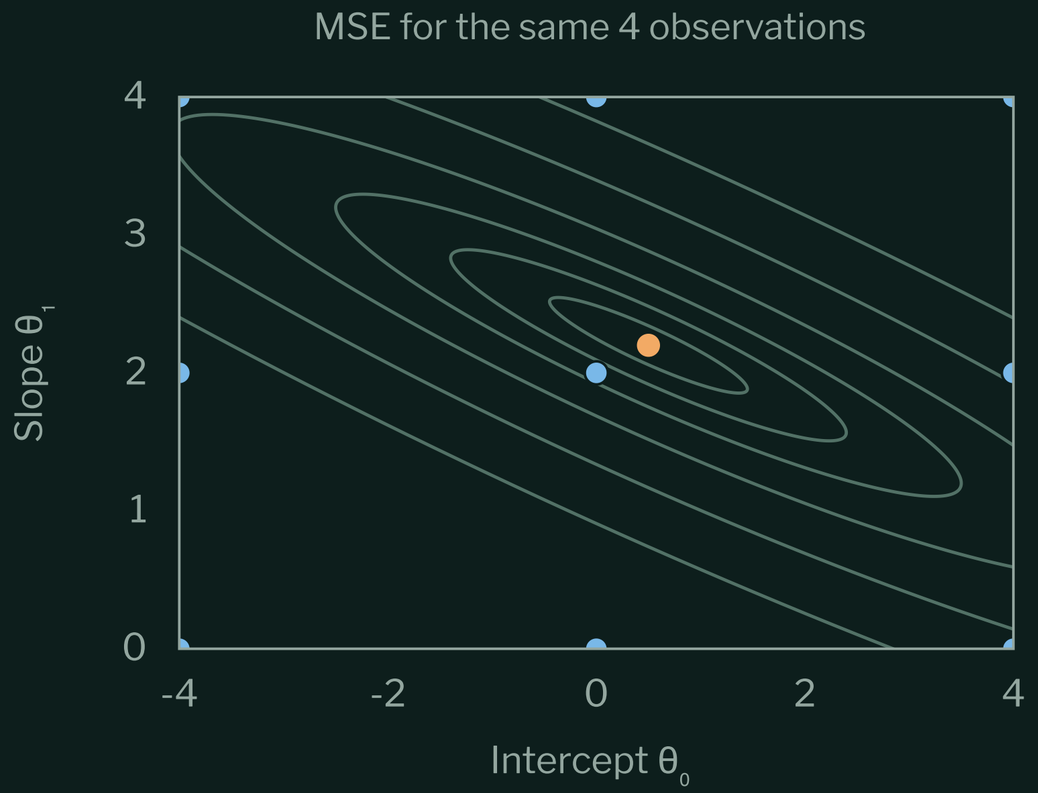
$\hat{y} = \theta_0 + \theta_1 x$ has interceptθ_0 and slope θ_1.

$\theta = (1, 2)$ means $\hat{y} = 1 + 2x$.

Constant predictor is same rule with slope 0: $\hat{y} = 6$ has coefficients $(6, 0)$.

Each point is one possible line

Method 2: Grid search



Intercepts $\in [-4, 4]$, slopes $\in [0, 4]$

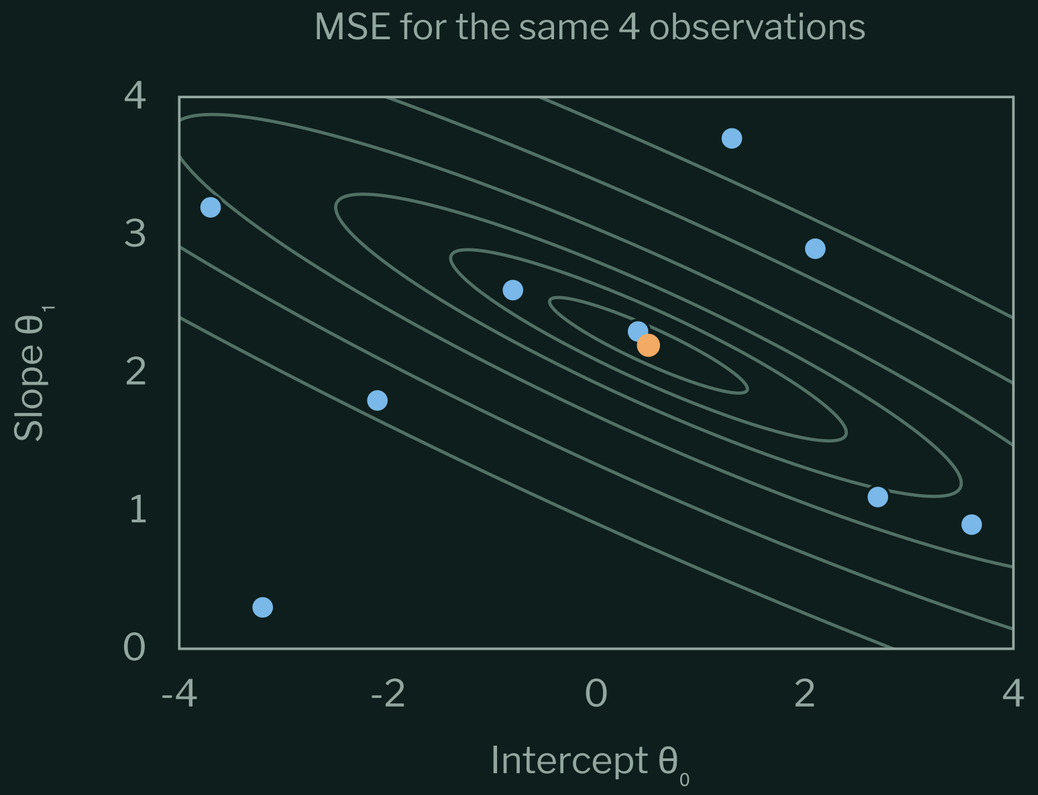
We cannot test every real-valued pair! Choose evenly spaced values

Try intercepts $-4, 0, 4$ and slopes $0, 2, 4$:
$3 \times 3 = 9$ tests.

Evaluate every grid point and keep the best

May not be OLS!

Method 3: Random search



Blue points: tested candidates

Orange: OLS minimum at (0.5,2.2)

Intercepts $\in [-4, 4]$, slopes $\in [0, 4]$

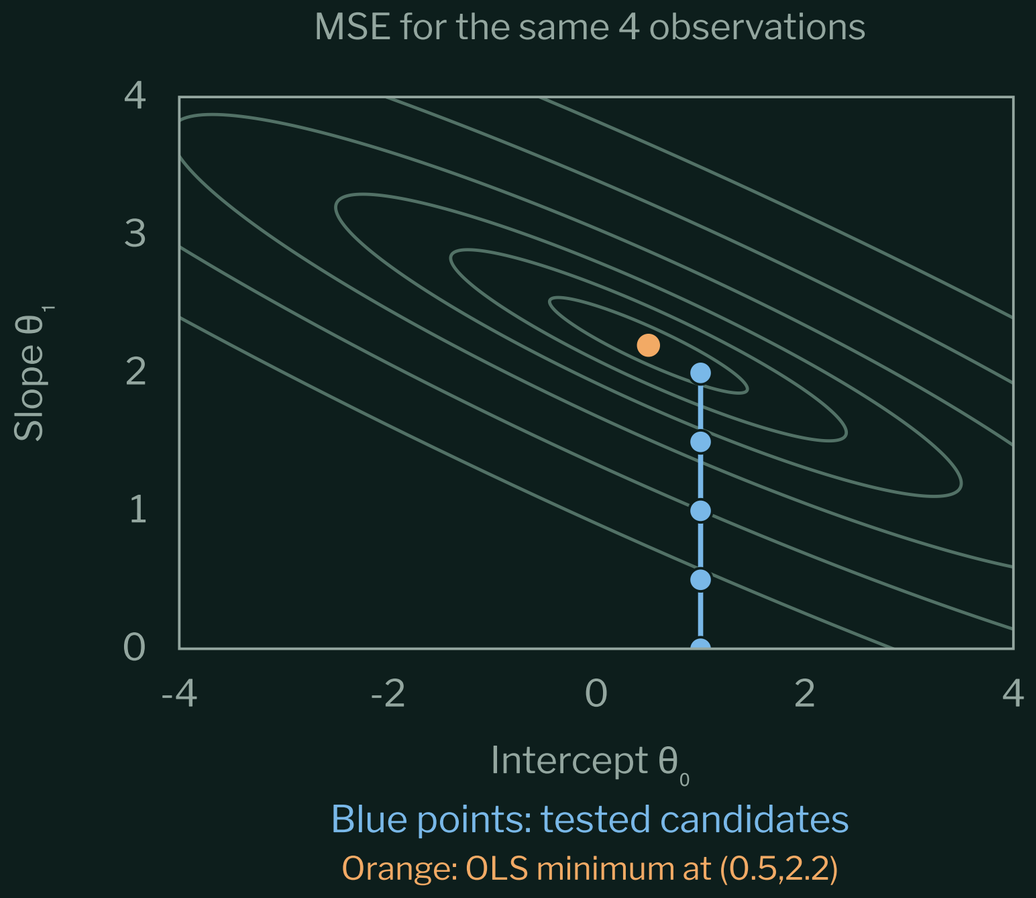
1. Pick an intercept and a slope at random

2. Calculate MSE, keep if it's better

3. Repeat until evaluation budget is used. Return best line tried

Each candidate drawn i.i.d.

Method 4: Hill climbing



Use current line to choose the next candidates (not i.i.d)

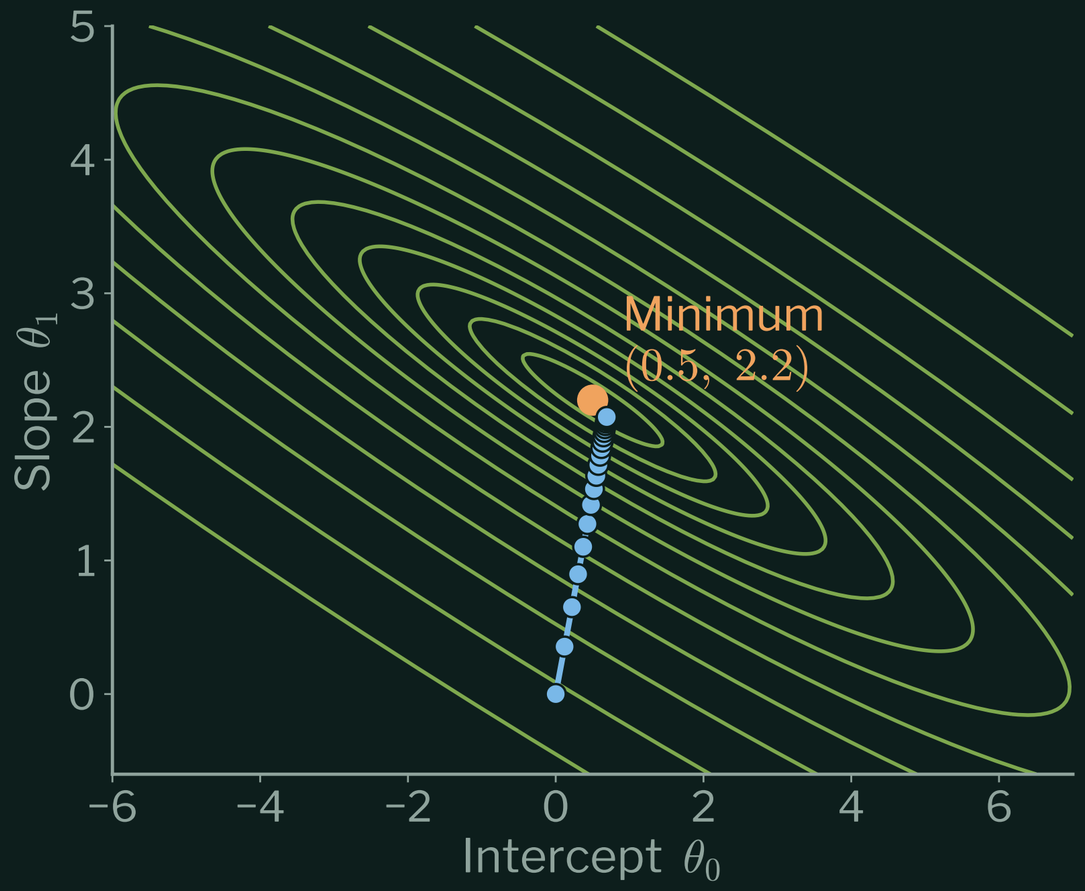
1. Start with e.g. $(1, 0)$.

2. Try increasing/decreasing 1 coef by step size $\gamma = 0.5$ (keep other coef fixed)

3. Evaluate + keep improvements

4. Repeat, stop if no improvements or after B

Method 6: Gradient descent



Calculate derivative for each coefficient at current line. Write these derivatives as $g_0, g_1, \ldots$.

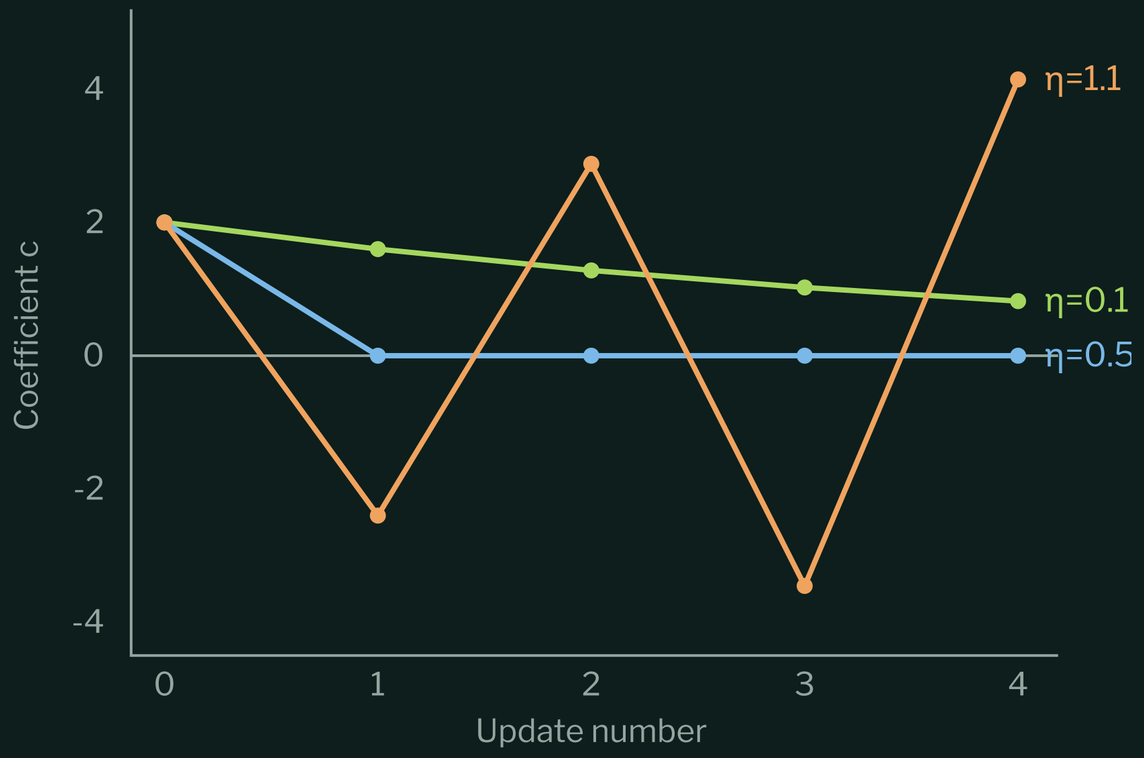
Positive $\implies$ small increase worsens it

Use $\theta_j^{\text{new}} = \theta_j - \eta g_j$, where $\eta > 0$ is learning rate controlling step size.

Calculate all derivatives, update coefficients together

Rinse, repeat

Gradient descent: step size



$L(c) = c^2$, minimum at $c = 0$ and its derivative is $2c$.

Update is $c^{\text{new}} = c - \eta(2c) = (1 - 2\eta)c$.

With $\eta = 0.1$, multiply by 0.8: $2 \to 1.6 \to 1.28$

With $\eta = 0.5$, multiply by 0; **done**

With $\eta = 1.1$, multiply by -1.2:
$2 \to -2.4 \to 2.88$

Classification

(via Logistic Regression)

What about binary targets?

Linear regression

Wait time in minutes

$$\hat{y} = b + wx$$

Binary classification

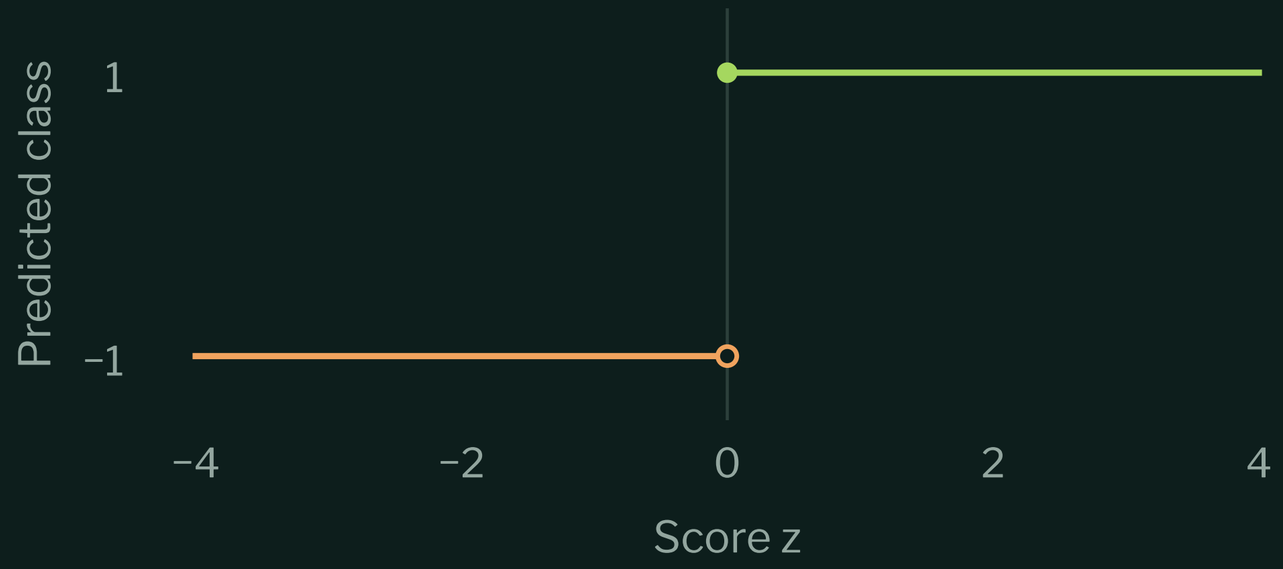
Spam or nonspam

$y = -1$: Nonspam

$y = +1$: Spam

Class prediction from sign

$$z = b + wx$$



$z < 0$

$\hat{y} = -1$

Nonspam

$z \geq 0$

$\hat{y} = +1$

Spam

Same class / different scores

Score close to zero

$$z = 0.1 \quad \Rightarrow \quad \hat{y} = +1$$

Score far from zero

$$z = 4 \quad \Rightarrow \quad \hat{y} = +1$$

Same class label + no **degree** of certainty

How do we find optimal coefficients?

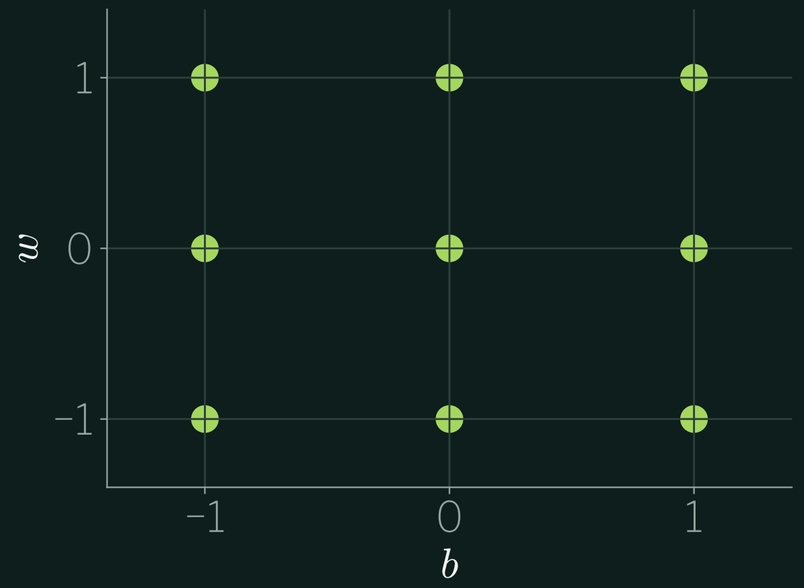
Candidate coefficients (b, w) $\longrightarrow$ Training predictions $\hat{y}_i = \mathbf{sign}(b + wx_i)$ $\longrightarrow$ Wrong predictions $R(b, w) = \frac{\text{Wrong labels}}{n}$

$$(b, w) \longrightarrow R(b, w) \longrightarrow \min_{b,w} R(b, w)$$

Search methods

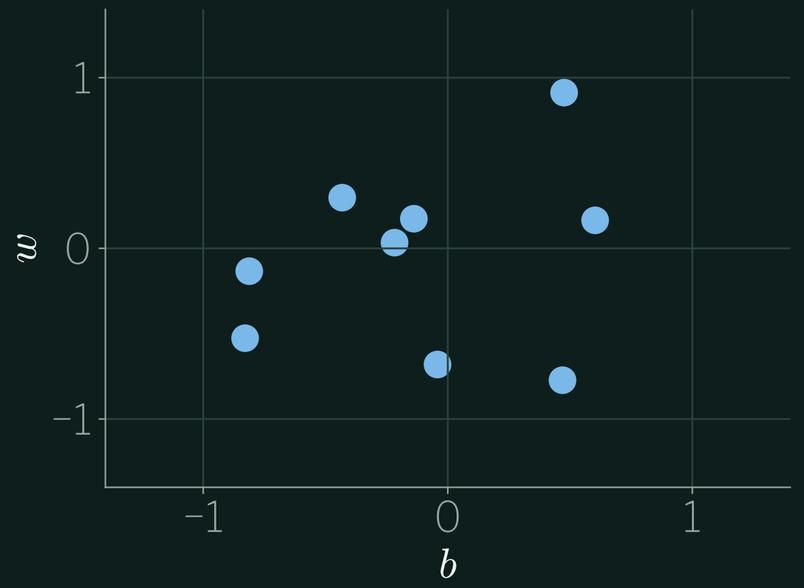
Grid search

$$b, w \in \{-1, 0, 1\}$$



Random search

$$-1 \leq b, w \leq 1$$



Keep coefficients with lowest measured training error

Can we use gradient descent?

Training error

$$R(b, w)$$

$$\hat{y}_i = \text{sign}(b + wx_i)$$

$\longrightarrow$

Required gradients

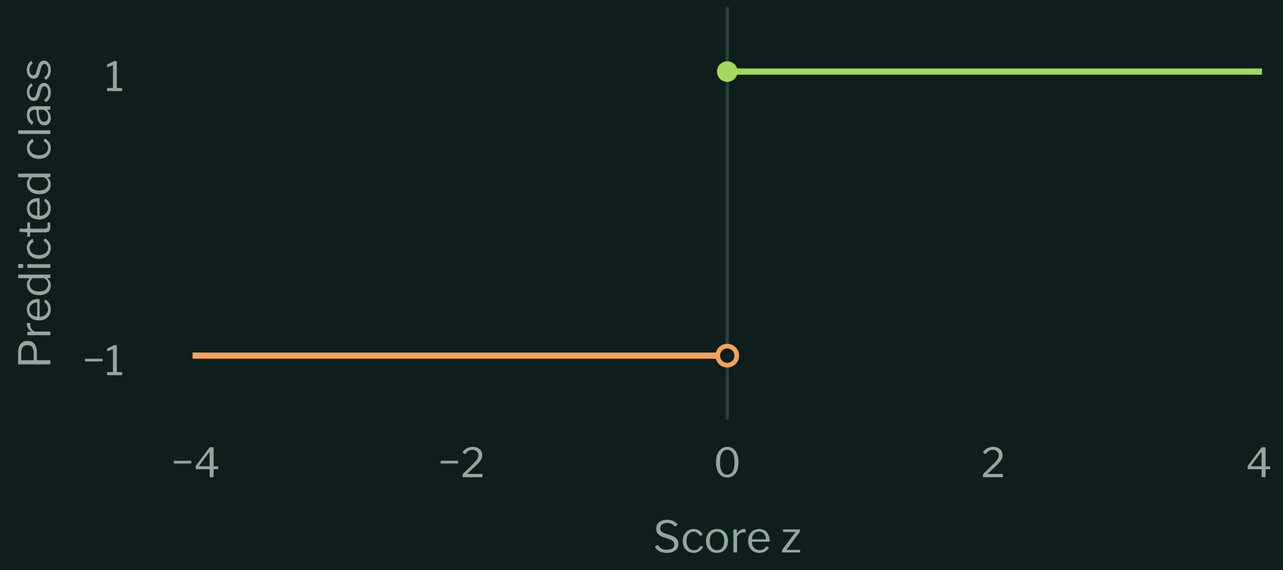
$$\frac{\partial R}{\partial b}, \qquad \frac{\partial R}{\partial w}$$

$$b^{\text{new}} = b - \eta \frac{\partial R}{\partial b}$$

$$w^{\text{new}} = w - \eta \frac{\partial R}{\partial w}$$

What happens when we differentiate sign?

$$f(z) = \text{sign}(z)$$



$$z \neq 0$$

$$f(z+h) = f(z)$$

$$f'(z) = \frac{f(z+h) - f(z)}{h} = 0$$

$$z = 0$$

$$f'(0) \text{ undefined}$$

Smooth probabilities from linear scores

$z = b + wx; p = \sigma(z)$, estimated spam **probability**

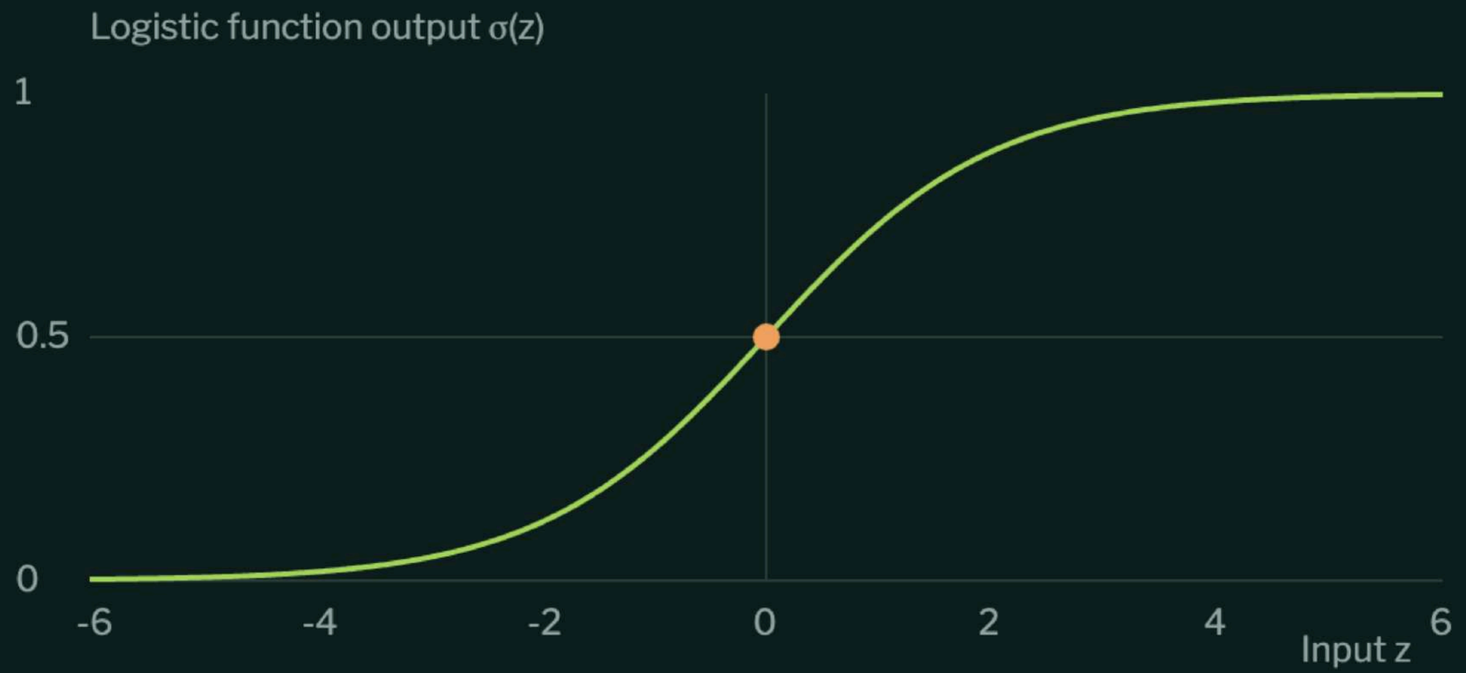
Function input z

0.0

$\sigma(z) = 1 / (1 + e^{-z})$

Fixed function
Selected input changes
Output between 0 and 1

Reset



e^{-z}	Denominator	Output σ(z)
1.000	**2.000**	**0.500**

Logistic function

$$p = \sigma(z) = \frac{1}{1+e^{-z}}$$

z	e^{-z}	p
-1	$e \approx 2.718$	0.269
0	1	0.5
1	$1/e \approx 0.368$	0.731

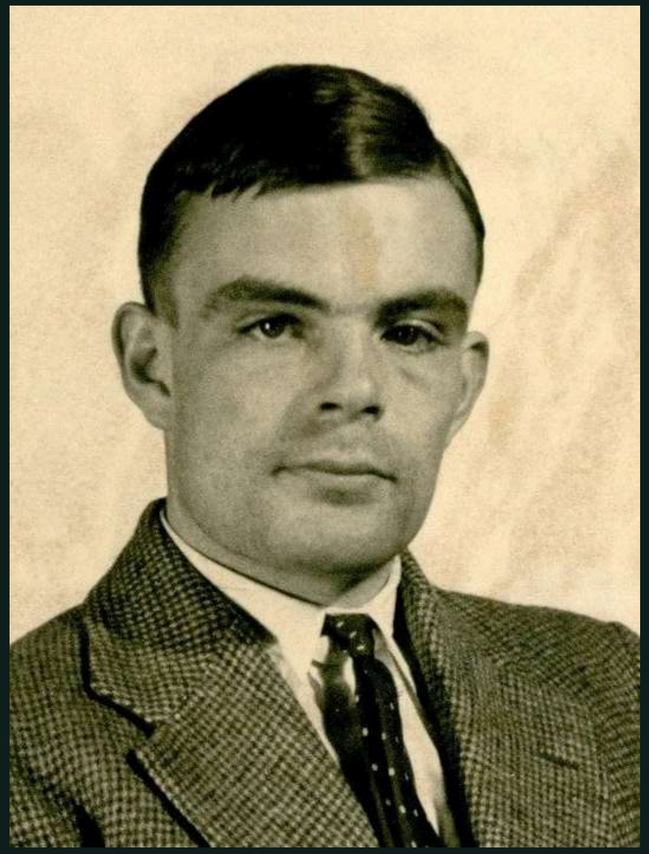
$$e^{-z} > 0 \quad \Longrightarrow \quad 0 < p < 1$$

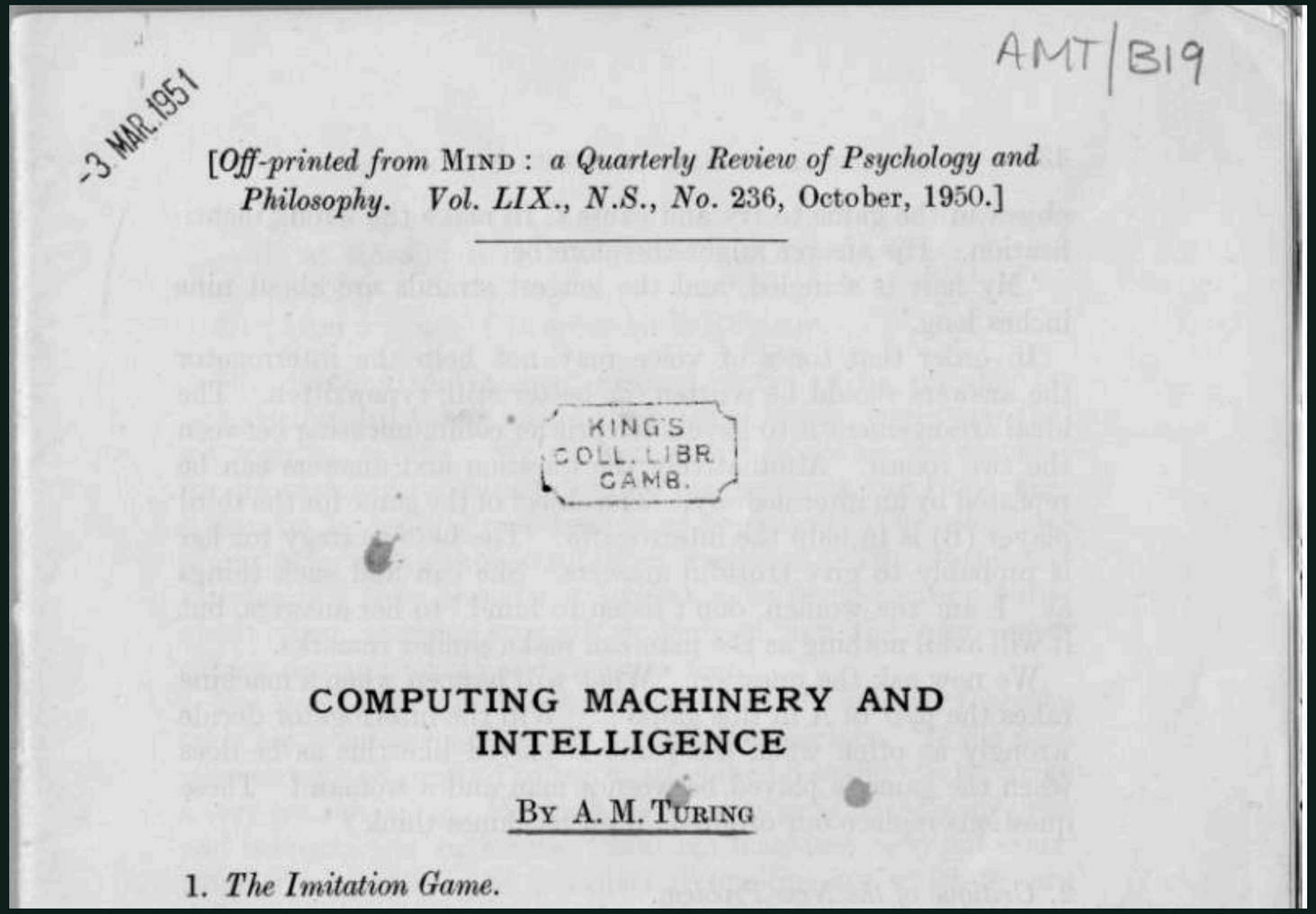
Quick aside

The Imitation Game



Alan Turing, 1950





-3 MAR 1951

AMT/B19

[*Off-printed from* MIND : *a Quarterly Review of Psychology and Philosophy. Vol. LIX., N.S., No.* 236, October, 1950.]

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COMPUTING MACHINERY AND INTELLIGENCE

BY A. M. TURING

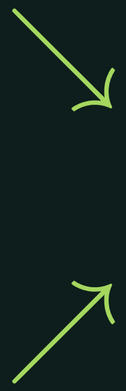
1. *The Imitation Game.*

Turing test as classification

Hidden participants
True labels y

Human

Machine



Text conversation
Input x



Human interrogator
Classifier



Predicted label $\hat{y}$
Human or machine

Classification error: Wrong predicted identity

Who plays Alan Turing?







Penguins/Penwings/Pingwings/Penlengs

POV: You're an Antarctic ecology researcher

How does changing sea ice affect penguin feeding?

Compare feeding

Across penguin species

→

Species labels

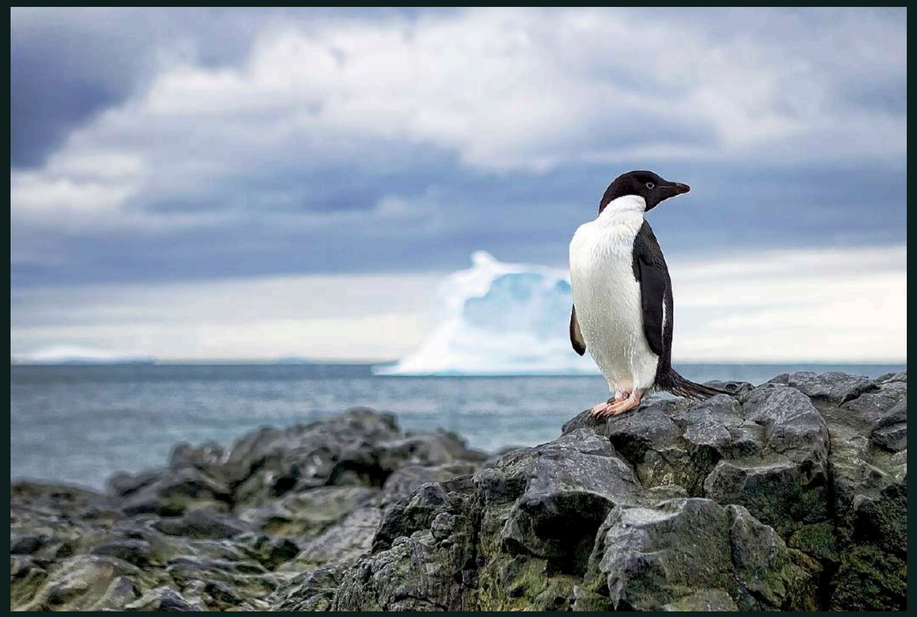
For each measurement

→

Missing species

Predict from body measurements

Species with different winter habitats



Adélie

Winter habitat: Pack ice

Christopher Michel, 2005; CC BY 2.0

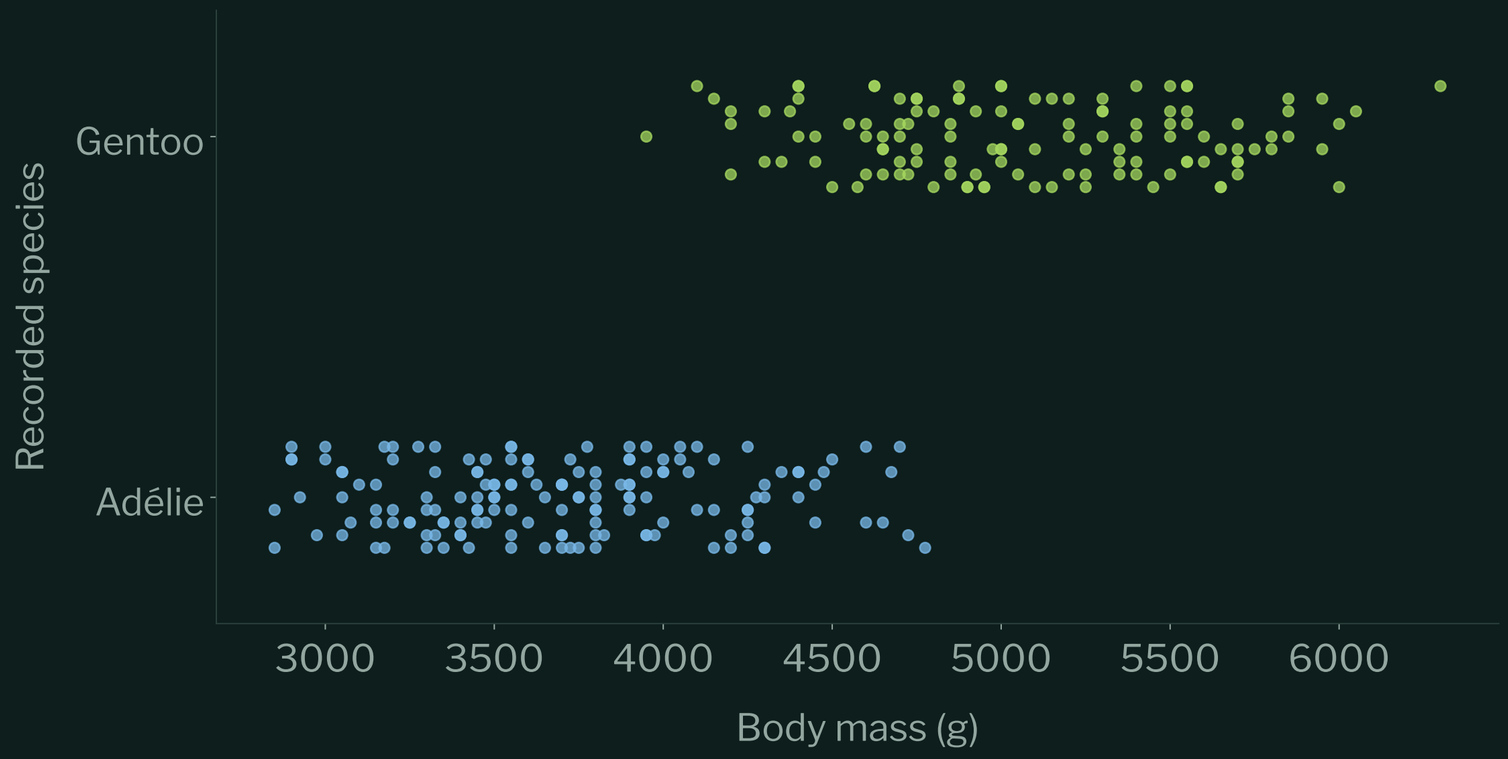


Gentoo

Winter habitat: Nearshore open water

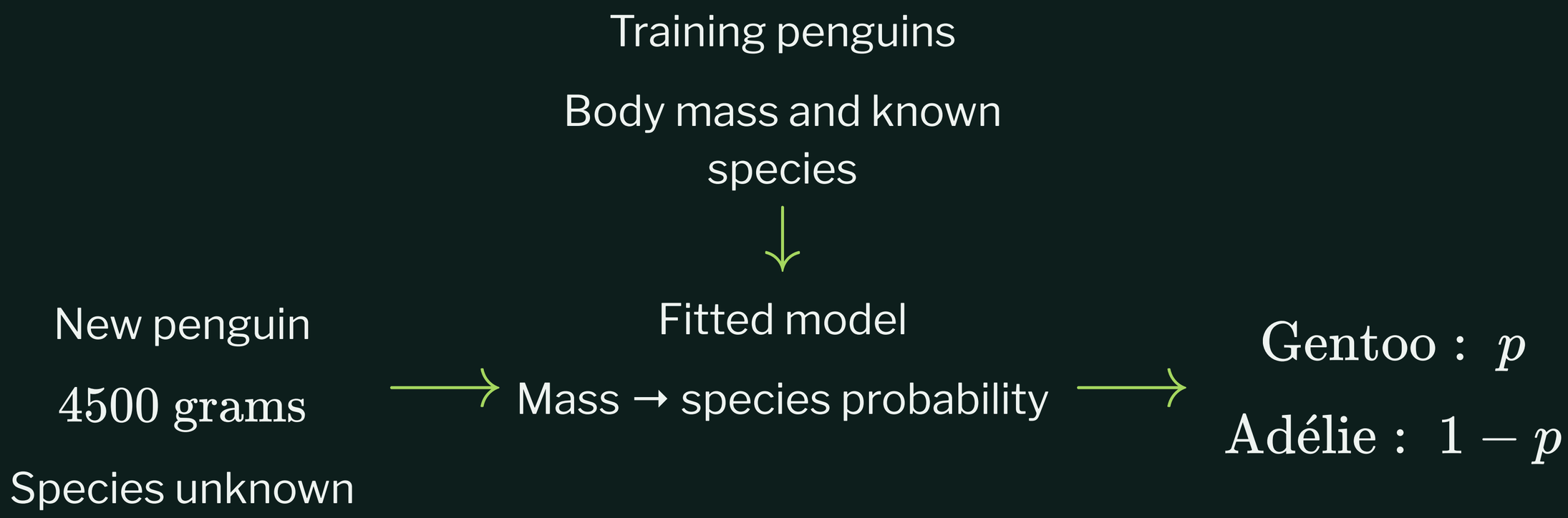
W. Bulach, 2005; CC BY-SA 4.0

Measured masses



151 Adélie and 123 Gentoo penguins

Predict species from body mass



Features and labels

$$i: \text{ Penguin row}$$

Measured mass → Feature

$$x_i = \frac{\text{Mass}_i - 4000}{1000}$$

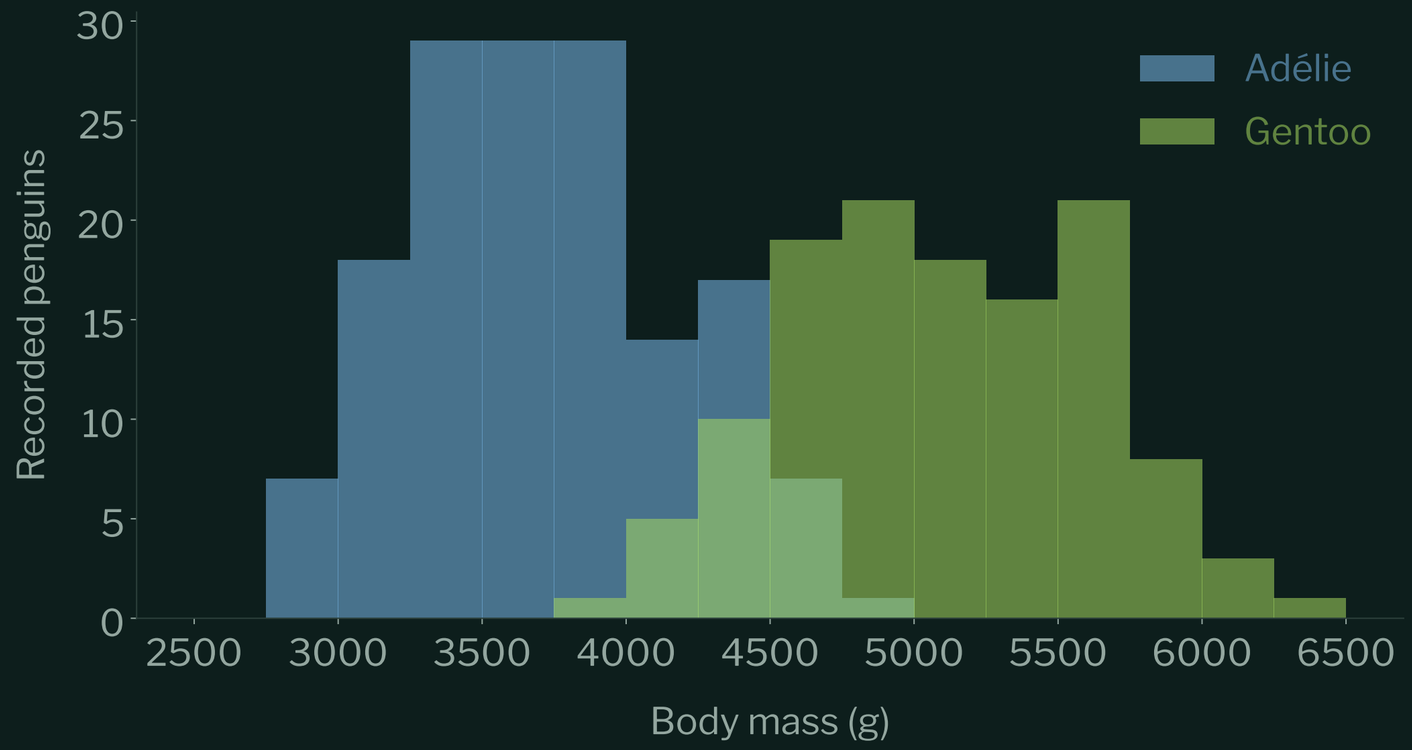
Mass in grams	x_i
3000	-1
4000	0
5000	1

Recorded species → Label

$$y_i \in \{0, 1\}$$

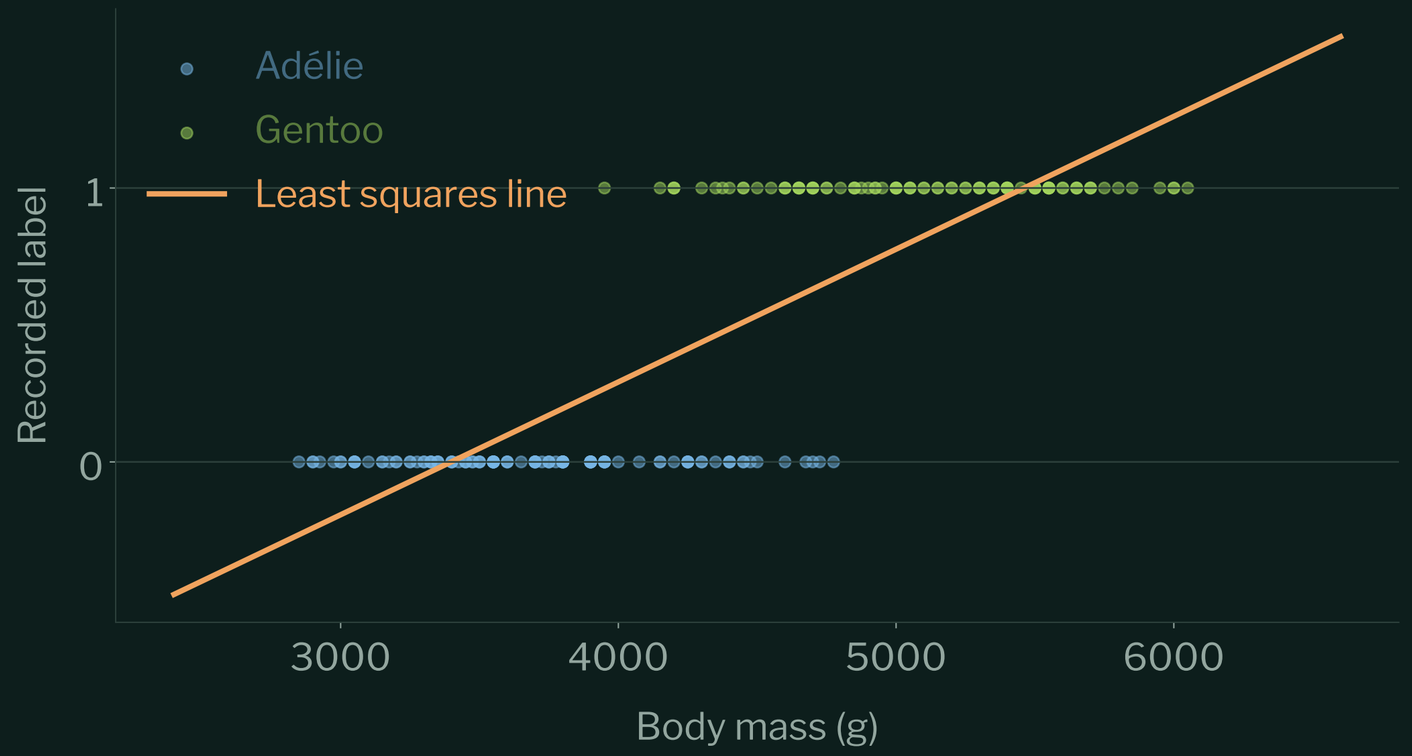
Species	y_i
Adélie	0
Gentoo	1

Overlapping body masses



Same mass → possible Adélie or Gentoo

Binary labels



$p \in [0, 1]$ $\hat{y}_{\text{OLS}}$ can be $\notin [0, 1]$

Score and probability

Mass feature

x

$\longrightarrow$

Linear score

$z = b + wx$

$z \in \mathbb{R}$

$\longrightarrow$

Gentoo probability

$p = \widehat{P}(Y = 1 \mid x)$

$0 < p < 1$

Recorded penguin

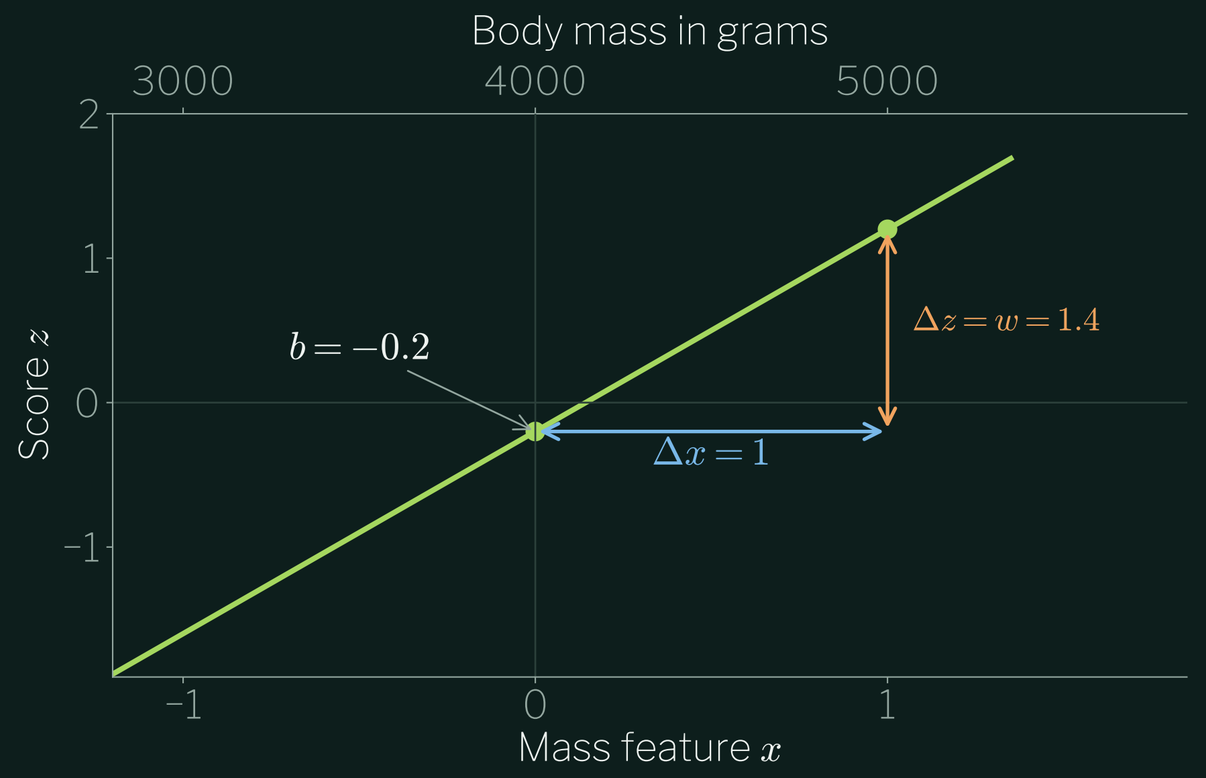
$y = 0 \text{ or } 1$

Species of new penguin

$Y \in \{0, 1\}$

Linear score

$$z = b + wx$$



Calculate score

Measured mass

4500 grams

$\longrightarrow$

Mass feature

$x = \frac{4500 - 4000}{1000} = 0.5$

$\longrightarrow$

Linear score

$z = -0.2 + 1.4(0.5)$

$z = 0.5$

$$b = -0.2, \qquad w = 1.4$$

Calculate species probabilities

$$4500 \text{ grams} \longrightarrow x = 0.5$$

Calculated score

$$z = 0.5$$

$\longrightarrow$

Logistic function

$$p = \frac{1}{1 + e^{-z}}$$

$$p = \frac{1}{1 + e^{-0.5}}$$

$\longrightarrow$

Gentoo

$$p \approx 0.622$$

Adélie

$$1 - p \approx 0.378$$

Odds from same prediction

$$4500 \text{ grams} \longrightarrow z = 0.5$$



$$\text{Odds} = \frac{p}{1-p} \approx \frac{0.622}{0.378} \approx 1.65$$

Gentoo probability $\approx 1.65\times$ Adélie probability

Score equals log odds

Probability

$$p = \frac{1}{1 + e^{-z}}$$

Complement

$$1 - p = 1 - \frac{1}{1 + e^{-z}}$$

$$1 - p = \frac{e^{-z}}{1 + e^{-z}}$$

$\longrightarrow$

Odds

$$\frac{p}{1 - p} = \frac{1}{1 + e^{-z}} \frac{1 + e^{-z}}{e^{-z}}$$

$$\frac{p}{1 - p} = \frac{1}{e^{-z}} = e^{z}$$

$\downarrow$

Log odds

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right) = \ln(e^{z}) = z = b + wx$$

Interpreting score

$$z = \ln\left(\frac{p}{1-p}\right)$$

Score z	Odds	Gentoo p	Higher probability
-0.5	0.61	0.378	Adélie
0	1	0.5	Equal
0.5	1.65	0.622	Gentoo

Changing coefficients

Intercept b

-0.2

Coefficient w

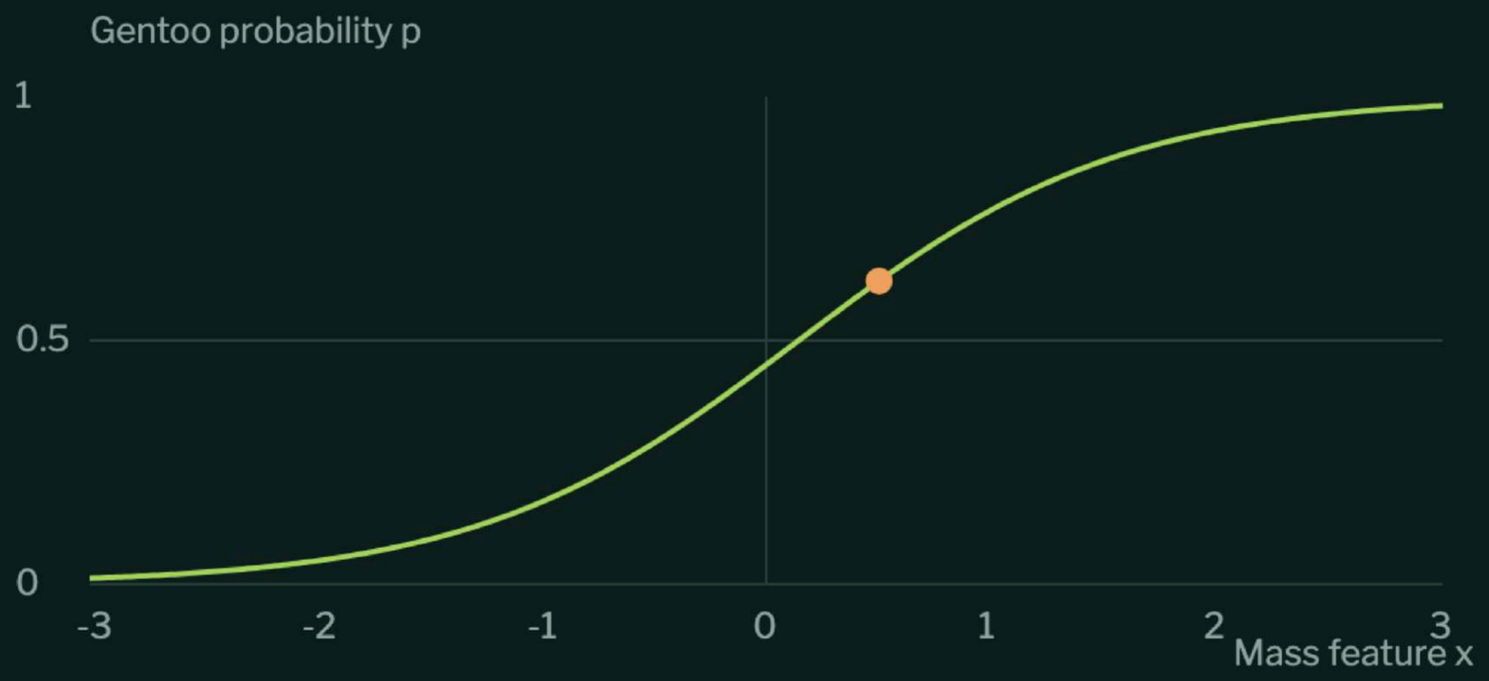
1.4

Mass feature x

0.5

x = (mass − 4000) / 1000
One unit: 1000 grams
p: estimated Gentoo probability

Reset



Score z	Odds	Probability p
0.50	**1.65**	**0.622**

Probability for recorded species

Same 4500-gram prediction

$$z = 0.5$$

$$p = \frac{1}{1 + e^{-0.5}} \approx 0.622$$

If recorded Gentoo $y = 1$

$$p \approx 0.622$$

If recorded Adélie $y = 0$

$$1 - p \approx 0.378$$

Log loss

Probability of recorded species

Gentoo $y_i = 1 \longrightarrow p_i$

Adélie $y_i = 0 \longrightarrow 1 - p_i$

Joint probability: all training penguins

$$\prod_{i=1}^{n} p_i^{y_i} (1 - p_i)^{1 - y_i}$$

$\longrightarrow$

Negative log: product $\rightarrow$ sum

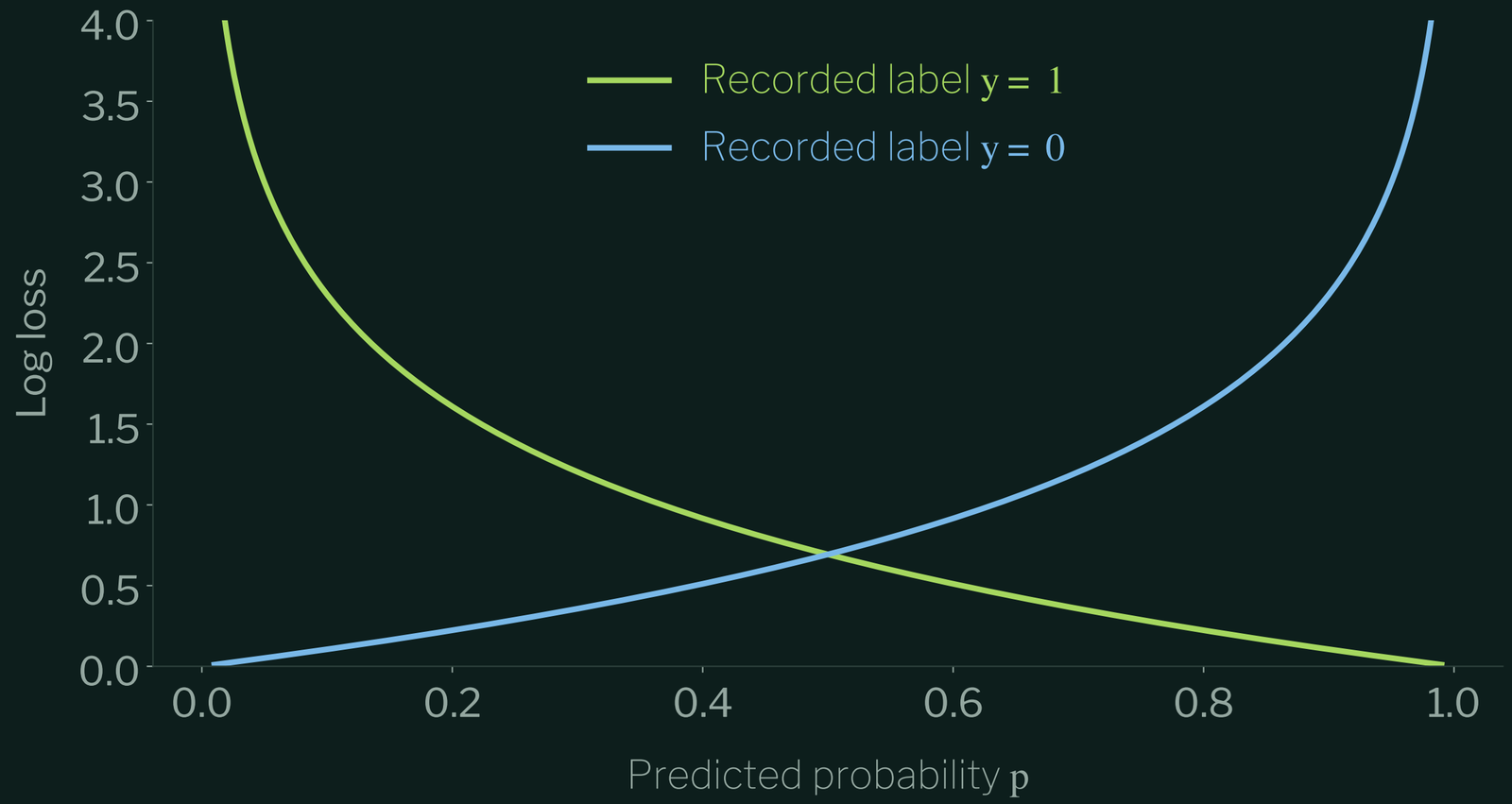
$$-\ln(\text{joint probability}) = \sum_{i=1}^{n} \ell_i$$

$\downarrow$

Average log loss

$$L = \frac{1}{n} \sum_{i=1}^{n} \ell_i = -\frac{1}{n} \sum_{i=1}^{n} \left[y_i \ln p_i + (1 - y_i) \ln(1 - p_i) \right]$$

Probability and loss



Loss increases with confidence in incorrect class

Same prediction, different recorded species

Same 4500-gram prediction

$$p \approx 0.622$$

If recorded Gentoo $y = 1$

$$p \approx 0.622$$

$$\ell = -\ln p \approx 0.474$$

If recorded Adélie $y = 0$

$$1 - p \approx 0.378$$

$$\ell = -\ln(1 - p) \approx 0.974$$

Calculate probability and loss

Mass feature → score → probability → loss

```
import math
b, w  # Score intercept; mass coefficient
x     # Centered mass in 1000 gram units
y     # Recorded label: 0 Adélie, 1 Gentoo
```

```
z = Python expression
p = Python expression
q = Python expression
loss = Python expression
```

Run checks | Reset

Python ready

Find coefficients

Smallest average training loss

$$\underset{b,w}{\text{minimize}}\ L(b, w)$$

No direct formula for best b, w → Gradient descent →

Need coefficient gradients

$$\frac{\partial L}{\partial b} \qquad \frac{\partial L}{\partial w}$$

Derivative of probability

$$p = \frac{1}{1 + e^{-z}} = (1 + e^{-z})^{-1}$$

Differentiate

$$\frac{dp}{dz} = -(1 + e^{-z})^{-2}(-e^{-z})$$

$$\frac{dp}{dz} = \frac{e^{-z}}{(1 + e^{-z})^2}$$

Simplify using probability

$$1 - p = \frac{e^{-z}}{1 + e^{-z}}$$

$$p(1 - p) = \frac{e^{-z}}{(1 + e^{-z})^2}$$

$$\frac{dp}{dz} = p(1 - p) = \frac{e^{-z}}{(1 + e^{-z})^2}$$

Derivative of loss

$$p = \frac{1}{1 + e^{-z}} \qquad \frac{dp}{dz} = p(1-p) = \frac{e^{-z}}{(1+e^{-z})^2}$$

Recorded Gentoo $y = 1$

$$\ell = -\ln p$$

$$\frac{d\ell}{dp} = -\frac{1}{p}$$

$$\frac{d\ell}{dz} = -\frac{1}{p}\, p(1-p)$$

$$= p - 1$$

Recorded Adélie $y = 0$

$$\ell = -\ln(1-p)$$

$$\frac{d\ell}{dp} = \frac{1}{1-p}$$

$$\frac{d\ell}{dz} = \frac{1}{1-p}\, p(1-p)$$

$$= p$$

Both labels

$$\frac{d\ell}{dz} = p - y = \frac{1}{1 + e^{-z}} - y$$

Coefficient gradients

$$z = b + wx \qquad \frac{d\ell}{dz} = p - y = \frac{1}{1 + e^{-z}} - y$$

Intercept b

$$\frac{\partial z}{\partial b} = 1$$

$$\frac{\partial \ell}{\partial b} = \frac{d\ell}{dz}\frac{\partial z}{\partial b} = (p - y) \times 1$$

$$\frac{\partial \ell}{\partial b} = \frac{1}{1 + e^{-z}} - y$$

Feature coefficient w

$$\frac{\partial z}{\partial w} = x$$

$$\frac{\partial \ell}{\partial w} = \frac{d\ell}{dz}\frac{\partial z}{\partial w} = (p - y) \times x$$

$$\frac{\partial \ell}{\partial w} = \left(\frac{1}{1 + e^{-z}} - y\right) x$$

Example: $p = 0.8,\ y = 1,\ x = 0.5$

$$\frac{\partial \ell}{\partial b} = -0.2 \qquad \frac{\partial \ell}{\partial w} = -0.1$$

Average gradients

$$L = \frac{1}{n}\sum_{i=1}^{n} \ell_i$$

$$z_i = b + wx_i \qquad p_i = \frac{1}{1 + e^{-z_i}}$$

Intercept

$$\frac{\partial \ell_i}{\partial b} = p_i - y_i$$

$$g_b = \frac{1}{n}\sum_{i=1}^{n}(p_i - y_i)$$

$$g_b = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{1}{1 + e^{-z_i}} - y_i\right)$$

Feature coefficient

$$\frac{\partial \ell_i}{\partial w} = (p_i - y_i)x_i$$

$$g_w = \frac{1}{n}\sum_{i=1}^{n}(p_i - y_i)x_i$$

$$g_w = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{1}{1 + e^{-z_i}} - y_i\right)x_i$$

g_b, g_w: Average training loss gradients

Gradient update

Current gradients

$g_b, \quad g_w$

Same current coefficients

→

Simultaneous update

$$b_{\text{new}} = b - \eta g_b$$

$$w_{\text{new}} = w - \eta g_w$$

→

New predictions

$$z_{\text{new}} = b_{\text{new}} + w_{\text{new}} x$$

$$p_{\text{new}} = \sigma(z_{\text{new}})$$

One gradient update

$$b = w = 0 \quad \Longrightarrow \quad p_i = 0.5 \qquad \eta = 0.3$$

x_i	y_i	$p_i - y_i$	$(p_i - y_i)x_i$
-1	0	0.5	-0.5
0	0	0.5	0
1	1	-0.5	-0.5
Mean		$g_b = \frac{1}{6}$	$g_w = -\frac{1}{3}$

$$b_{\text{new}} = 0 - 0.3\left(\tfrac{1}{6}\right) = -0.05 \qquad w_{\text{new}} = 0 - 0.3\left(-\tfrac{1}{3}\right) = 0.1$$

Calculate gradient update

Probability error → simultaneous update

```
b, w  # Current score coefficients
x     # Centered mass in 1000 gram units
p     # Predicted Gentoo probability
y     # Recorded label: 0 Adélie, 1 Gentoo
eta   # Learning rate; simultaneous updates
```

error = Python expression

db = Python expression

dw = Python expression

b_next = Python expression

w_next = Python expression

Run checks Reset

Python ready

Fit penguin probabilities

Split observations

191 training penguins

83 test penguins

→ **Fit on training rows**

$b = w = 0$

1000 updates

$\eta = 0.08$

→ **Evaluate test predictions**

Coefficients fixed after training

83 held-out species labels

Log-loss coefficients: No general formula for best fit

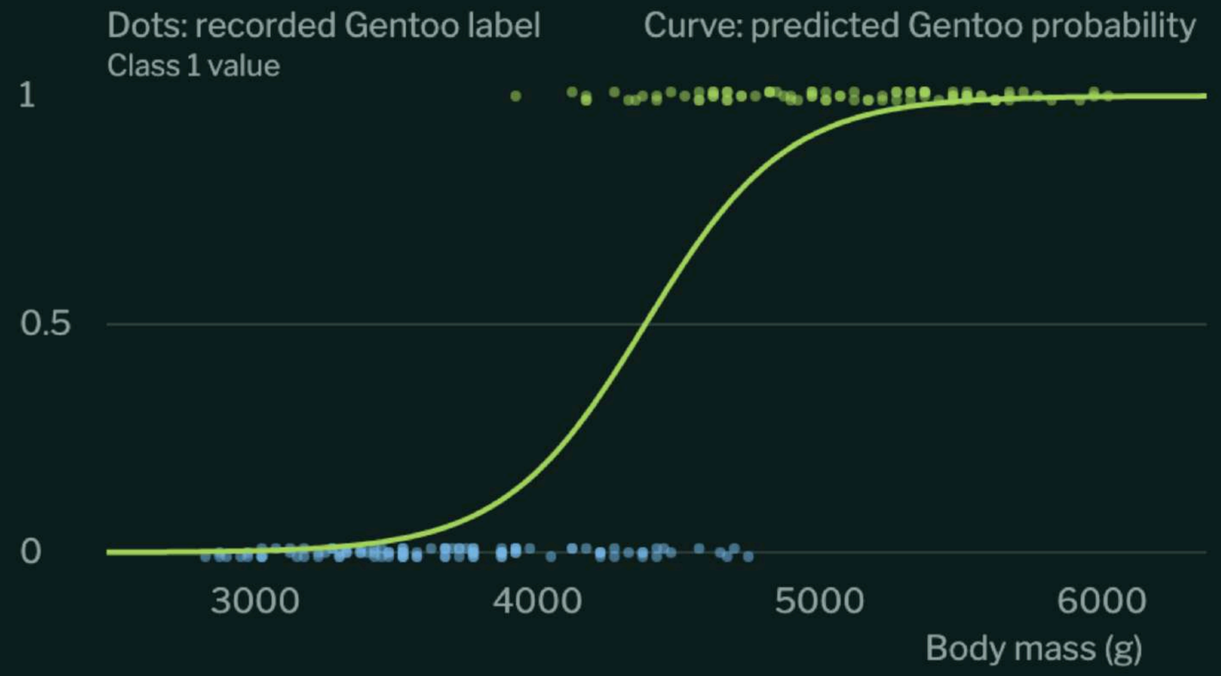
Computable gradients → Iterative coefficient updates

Fitting probabilities

Play

Reset

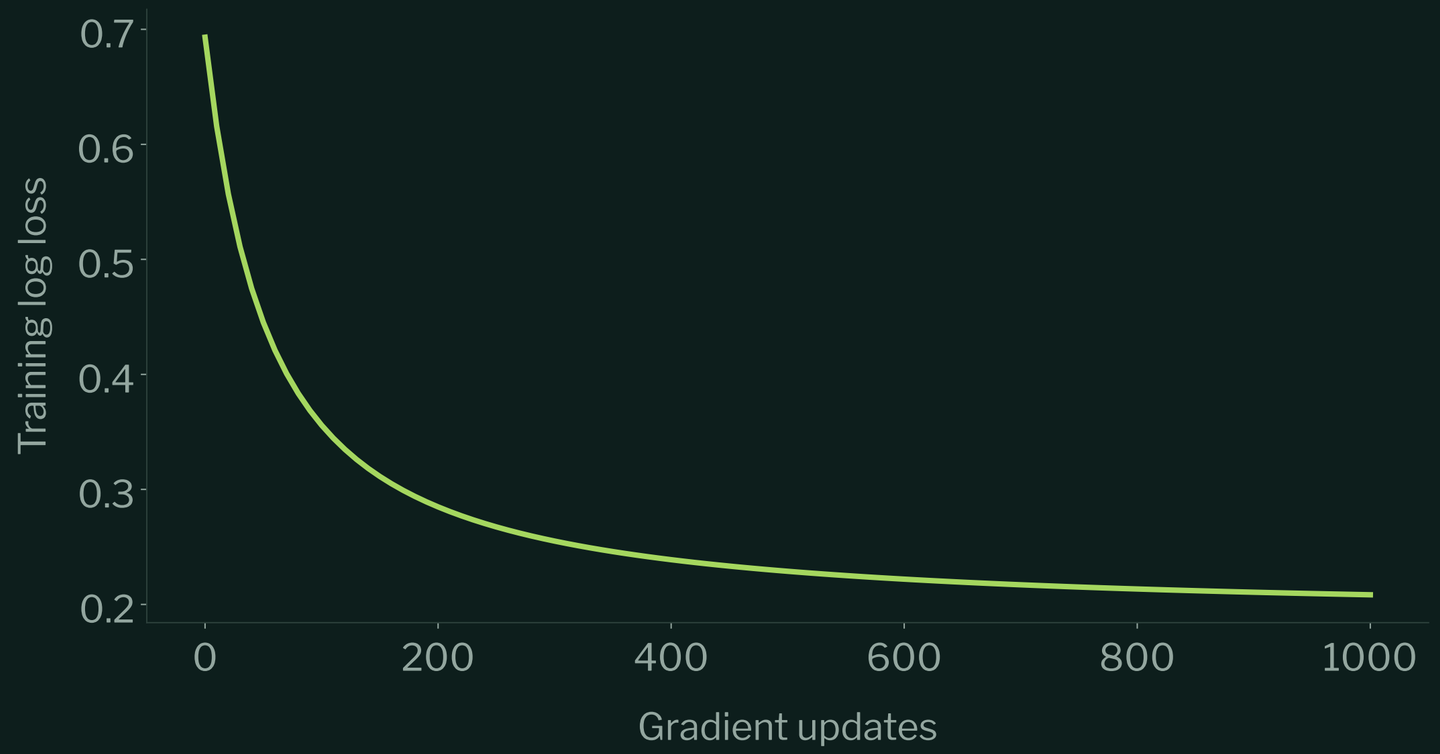
Update 1000





Intercept b **-1.639** Coefficient w **4.002** Training log loss **0.208**

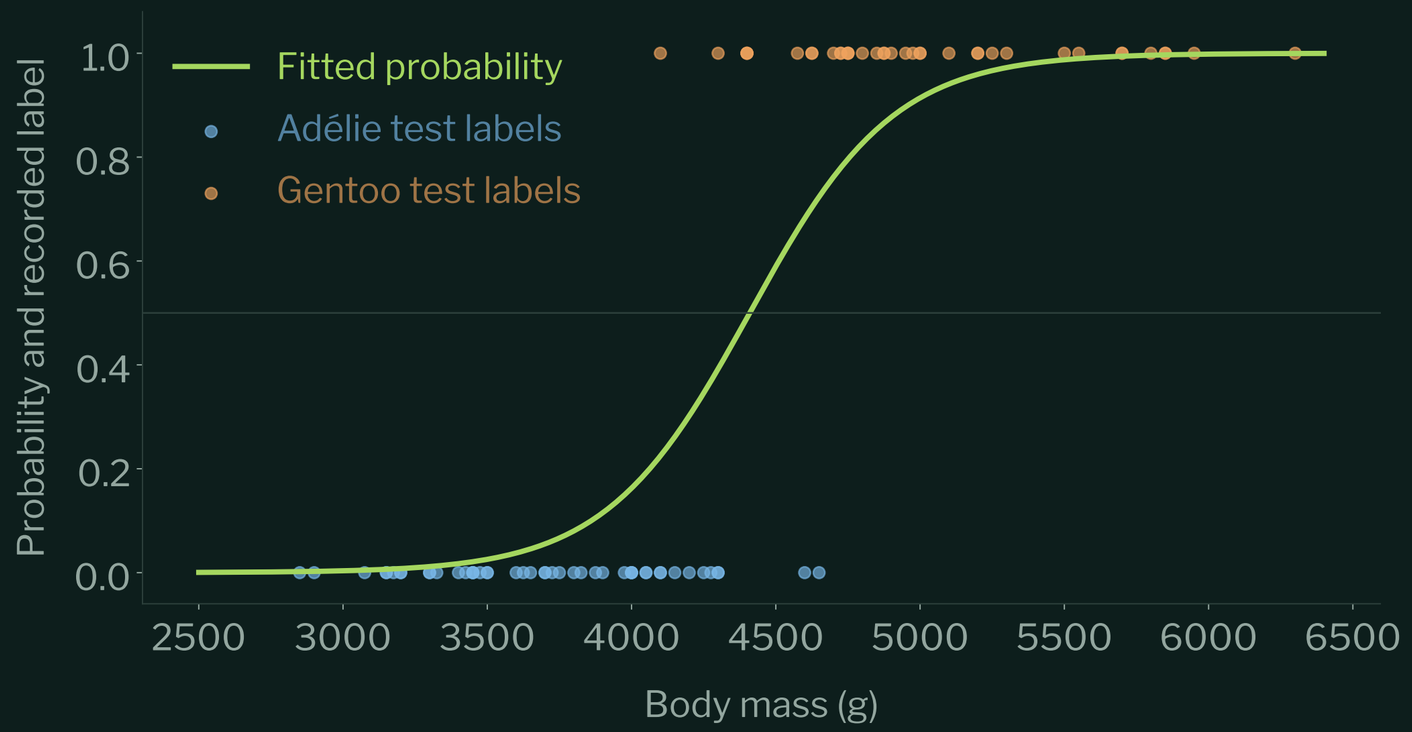
Training loss



191 training rows; average loss for recorded species

Same rows at every update; smaller loss means better training fit

Probability after **1000** updates



$$b \approx -1.639, \quad w \approx 4.002$$

Curve: Fitted on training data probabilities; dots: 83 held-out labels

Class prediction errors

$$\hat{y} = \begin{cases} 1 \text{ (Gentoo)} & p \geq 0.5 \\ 0 \text{ (Adélie)} & p < 0.5 \end{cases}$$

Correct species

$$\hat{y} = y$$

Error 0

Wrong species

$$\hat{y} \neq y$$

Error 1

Test penguins	Error rate	Accuracy
83	$\frac{7}{83} \approx 8.4\%$	$\frac{76}{83} \approx 91.6\%$

Which predictions are wrong?

Positive class: Gentoo

	Recorded Gentoo	**Recorded Adélie**
Predicted Gentoo	True positive (TP) 32	False positive (FP) 2
Predicted Adélie	False negative (FN) 5	True negative (TN) 44

$$\text{Correct} = \text{TP} + \text{TN} = 76 \qquad \text{Wrong} = \text{FP} + \text{FN} = 7$$

Precision and recall

Gentoo predictions on same 83 test penguins

Precision
Among predicted Gentoo

$$\frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\frac{32}{32 + 2} \approx 94.1\%$$

Recall
Among recorded Gentoo

$$\frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\frac{32}{32 + 5} \approx 86.5\%$$

Logistic regression takeaways

Mass feature

$$x$$

$\longrightarrow$

Linear score

$$z = b + wx$$

$\longrightarrow$

Gentoo probability

$$p = \frac{1}{1 + e^{-z}}$$

Fit coefficients

Recorded training species

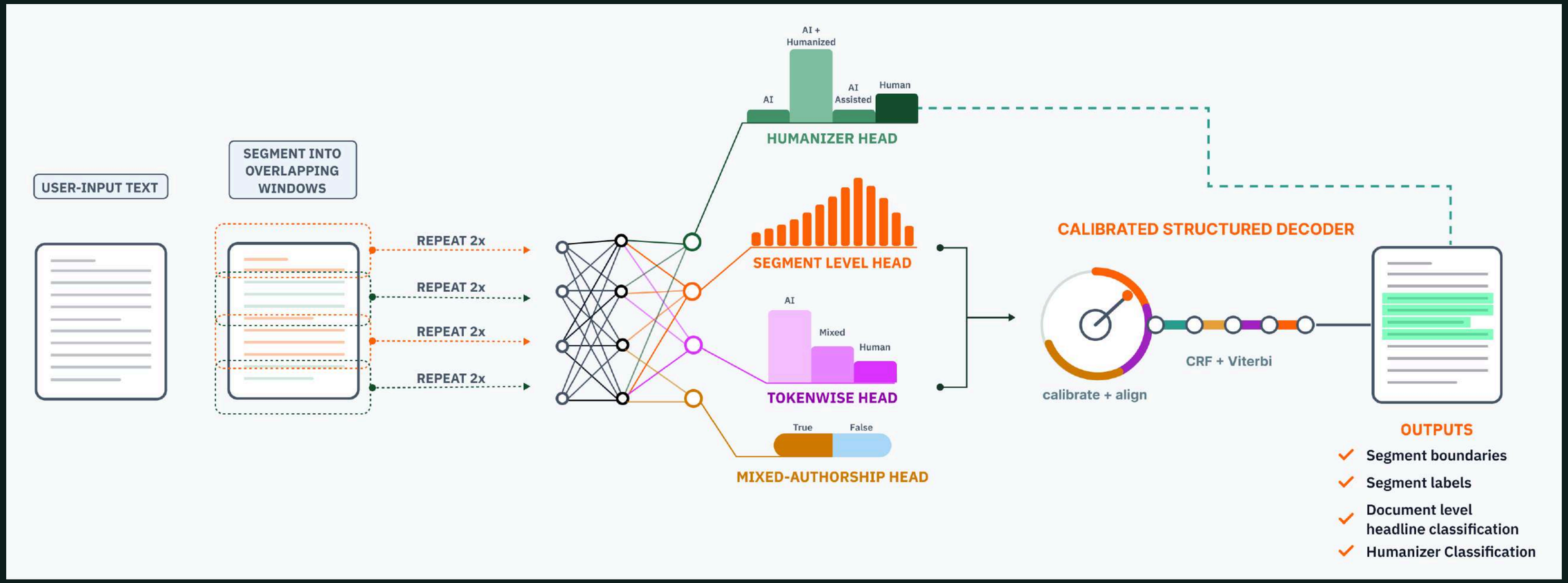
$\text{Log loss} \longrightarrow \text{Gradient descent}$

Predict new species

$p \geq 0.5 \Longrightarrow \text{Gentoo}$

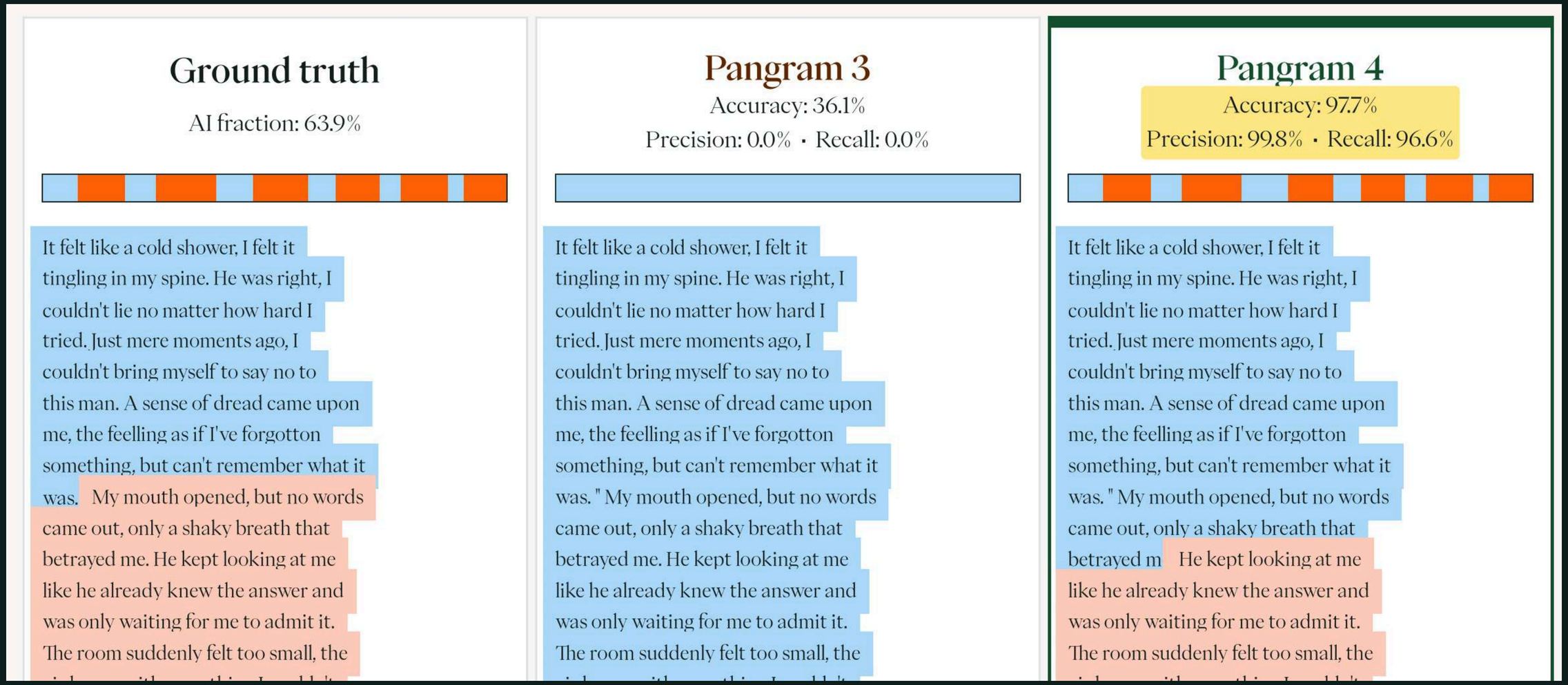
Check errors on test penguins

The "Imitation Game" Classification in 2026



Writing ⟶ Features ⟶ Authorship scores

Errors within mixed writing



Human AI-generated

Thinking Machines	Inkling	0.190%
Anthropic	Claude	0.195%
NVIDIA	Nemotron	0.310%
Alibaba Cloud	Qwen	0.315%
OpenAI	GPT	0.316%
Tencent	Hunyuan	0.330%
DeepSeek	V4	0.388%
Moonshot AI	Kimi	0.395%
Mistral AI	Mistral	0.400%
Google	Gemma	0.430%
Z.ai	GLM	0.470%
Google	Gemini	0.498%
Meta	Llama	0.550%
xAI	Grok	0.605%